


THE UNIVERSITY OF WESTERN AUSTRALIA

# CITS4402 Computer Vision

## Project Report

### Face Detection and Matching

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# 1. Introduction

This report describes the face detection and identity clustering system implemented for the CITS4402 project. The system processes 640×480 colour images (1–4 faces), detects faces, localises three landmarks (right eye, left eye, nose tip), aligns each face to a 125×125 crop, and clusters faces across a folder into unique identities. The pipeline uses MediaPipe BlazeFace [2] and FaceLandmarker [6] when available, with a fully self-contained fallback built on OpenCV [3] Haar cascades [10] and the Hough circle transform [11] covered in Lab 04. For clustering, 128-D dlib ResNet embeddings [5] (via the `face_recognition` library) feed DBSCAN [4]. The table below shows the headline measured outcomes from the naive starting point to the final implementation; subsequent sections detail each task's development.

## 1.1 Baseline vs Final (measured on the 12 sample images)

| Metric | Naive baseline | Final implementation |
|---|---|---|
| Face detection | 11/12 images (4.jpg missed) | 12/12 images, all counts correct |
| Landmark detection | 61% both-eyes (17/28 faces) | 100% both-eyes (26/26 faces) |
| Landmarks inside box | not measured | 100% RE, 100% LE, 100% nose |
| Alignment success | 61%, no similarity transform | 100%, exact targets (0 px residual) |
| Identity clustering | not implemented | eps=0.38, 6 clusters from 26 faces |

# 2. Face Detection (Task 2)

## 2.1 Development Iterations

| Version | Change | Outcome |
|---|---|---|
| v1 (baseline) | Haar cascade [10] (`frontalface_default`) only, no skin filter | 11/12 images; 4.jpg missed entirely |
| v2 | Added MediaPipe BlazeFace [2] (short-range + full-range) as primary detector; Haar retained as fallback | 12/12 images, but false positives: 5 boxes on 11.jpg (small body region), 1 face missed on 12.jpg (Ross failed cascade re-verification) |
| v3 | Added skin-coverage bypass for verification (skin $\geq$ 75% AND box $\geq$ 60 px); cap result at 4 (spec maximum) | Fixed 12.jpg (Ross retained) and 11.jpg (small body box dropped); 12/12 correct |
| v4 | Replaced YCrCb-only skin filter with three-rule majority vote (YCrCb + HSV wrap-around + normalised-RGB) | More tolerant of unusual lighting/darker skin without losing FP rejection |
| v5 (final) | CLAHE [12] replaces `equalizeHist` for the Haar-fallback grayscale input | 12/12 maintained; better local detail for the Haar fallback path |

## 2.2 Final Method

Primary: MediaPipe BlazeFace [2] short-range (230 KB) + full-range (1.1 MB) TFLite models, run on the original and horizontally-flipped image. Results merged with IoU-NMS (threshold 0.28, containment $>$ 72%). Each box expanded 20% up, 10% left/right, 15% down to capture the full head. Capped at 4 by ranking on confidence + skin-coverage score. Fallback when MediaPipe is unavailable: `frontalface_default` + `alt2` Haar cascades [10], 28% upward expansion, ROI re-verification, profile cascade as last resort.

## 2.3 Skin-Colour Segmentation (Mandatory)

Three-rule majority vote (Lab 05 technique): a pixel is skin when $\geq 2$ of three rules agree — YCrCb (Cr $\in [133, 183]$, Cb $\in [75, 135]$), HSV (H $\in [0, 28] \cup [160, 179]$), normalised RGB (R $>$ G $- 5$, R $>$ B, R $> 70$). Morphological open + close (5×5 ellipse) before the 20% pixel-ratio threshold.

## 2.4 Alternatives Considered

- dlib HOG face detector: lower recall on non-frontal faces vs BlazeFace.
- OpenCV YuNet: needs a separate ONNX file not bundled with `opencv-python`.

## 3. Facial Landmark Detection (Task 3)

### 3.1 Development Iterations

| Version | Change | Outcome |
|---|---|---|
| v1 (baseline) | `haarcascade_eye.xml` [10] only | 61% both-eyes (17/28); fails on tilted, glasses, occluded faces |
| v2 | Hough circle transform [11] with Lab 04 edge-overlap scoring on upper 58% ROI; `eye_tree_eyeglasses` and directional eye cascades as chained fallbacks; nose tip via intensity minimum of Gaussian-blurred patch | 100% both-eyes on all 26 faces |
| v3 | Added MediaPipe FaceLandmarker [6] (iris landmarks 468/473, nose landmark 1) as primary path | 100% maintained; sub-pixel iris precision |
| v4 | Per-eye position validation: reject MediaPipe landmarks where any single eye is outside the top 8–60% of the face box, or eyes differ in y by $>$ 20% face height | Caught a Chandler-in-7.jpg case where MediaPipe placed the right eye on the forehead, causing a badly tilted alignment |
| v5 (final) | Integral-projection profile + symmetry-mirror fallback tiers; CLAHE [12] preprocessing | 100% even with MediaPipe disabled (verified by forcing the landmarker off) |

### 3.2 Final Method

Seven-tier fallback chain (each tier only runs if the previous failed): (1) MediaPipe FaceLandmarker on a padded ROI; (2) Hough circles with Lab 04 edge-overlap scoring; (3) `haarcascade_eye_tree_eyeglasses`; (4) directional eye cascades; (5) integral-projection profile (darkest row in the upper face); (6) symmetry mirror if one eye is found; (7) proportional geometric estimate. Nose tip always from intensity minimum in the central 30–70% $\times$ 50–72% region of the face ROI.

### 3.3 Alternatives Considered

- dlib 68-point shape predictor: 68 MB model file, too large for submission.

## 4. Facial Alignment (Task 4)

### 4.1 Development Iterations

| Version | Change | Outcome |
|---|---|---|
| v1 (baseline) | `getRotationMatrix2D` for rotation, then separate crop + resize. Does not use a similarity transform; landmarks not at target positions | 61% success (tied to landmark failures) |
| v2 | `cv2.estimateAffinePartial2D` with LMEDS on all three landmarks | 100% success but visible tilt artefact when nose constraint is geometrically inconsistent with eyes (e.g., Monica in 7.jpg, eyes both at y=133 but face aligned with $\sim$20° rotation) |
| v3 | Analytical 2-point similarity transform from the two eye points only | 100%, no tilt; eyes exact at (40,40) and (85,40); but nose drifted from (63,70) — flagged by the lab facilitator |
| v4 | Least-squares similarity transform over all three landmarks (closed form via `np.linalg.lstsq` on the 6-equation 4-unknown system) | 100%; reduced tilt vs LMEDS but still measured max 9.5 px residual because a 4-DOF model cannot exactly fit 3 arbitrary point pairs |
| v5 (final) | `cv2.getAffineTransform` [3] with all three landmarks | 100%; verified 0 px residual on all 26 faces — every landmark lands precisely at its target coordinate |

### 4.2 Final Method

`cv2.getAffineTransform` takes exactly 3 source$\rightarrow$target pairs and returns the unique 6-DOF affine matrix that maps them exactly. A single `cv2.warpAffine` call with output size (125, 125) simultaneously aligns, crops, resizes, and (where needed) lightly stretches the face so that right_eye $\rightarrow (40, 40)$, left_eye $\rightarrow (85, 40)$, and nose_tip $\rightarrow (63, 70)$ are all exact. `borderMode=BORDER_REPLICATE` fills out-of-bounds pixels, preventing black borders. Landmark dots on the corner thumbnail are drawn

at the fixed target coordinates and never move between images.

## 5. Identity Clustering (Task 5)

### 5.1 Development Iterations

| Version | Change | Outcome |
|---|---|---|
| v1 (baseline) | Not implemented in the baseline code | — |
| v2 | 128-D dlib ResNet embeddings [5] via `face_recognition` (explicitly permitted by spec); DBSCAN [4] via scikit-learn [8] with `min_samples=1` and Euclidean distance | Clusters identities meaningfully; see eps sweep below |
| v3 | Singleton-merge post-processing: merge isolated one-face clusters into the nearest compatible cluster if distance $< 0.515$ AND no shared source image | Repairs cases where one image of a person has unusual pose/lighting |
| v4 | Upgraded the LBP fallback descriptor from basic 256-bin 5×5 to LBP-u2 [7] 7×7 multi-scale following Ahonen et al. [1]: 59 uniform-pattern bins × 49 cells × 2 radii = 5782-D L2-normalised | LBP fallback peak silhouette [9]: 0.04 → 0.11 ($\approx 3\times$ improvement) |

### 5.2 DBSCAN eps Sweep (final code, 26 faces)

| eps | Clusters | Silhouette [9] |
|---|---|---|
| 0.35 | 10 | 0.1408 |
| **0.38** | **6** | **0.1021 ← selected** |
| 0.40 | 4 | 0.0954 |
| 0.43 | 1 | 0.0000 |
| 0.45 | 1 | 0.0000 |

eps=0.38 was selected because 6 clusters is closest to the 5-identity ground truth (the 5 sample-image characters), with one slight over-split for a character whose appearance varies significantly between images. eps=0.35 has marginally higher silhouette (0.14) but over-splits to 10 clusters. eps=0.40 collapses two distinct identities. `min_samples=1` ensures every face joins a cluster (no noise points).

### 5.3 Alternatives Considered

- k-means: cannot be used — number of identities is unknown a priori.
- Agglomerative clustering: rejected because DBSCAN is explicitly named in the spec.

## 6. Final Results

Measured on the 12 sample images (26 detected faces, all face counts match ground truth):

- Face detection: 12/12 images correct; avg detection ≈ 39 ms (MediaPipe path).
- Landmark detection: 100% both eyes; 100% RE, 100% LE, 100% nose inside the face box.
- Alignment: 100% success; 0 px residual on every landmark; all outputs exactly 125×125.
- Clustering: 6 DBSCAN clusters from 26 faces (eps=0.38, silhouette = 0.10).

**Known limitations.** (1) MediaPipe model files (≈ 4.9 MB total) must accompany the script — the Haar/Hough fallback covers their absence automatically. (2) `face_recognition` (dlib) must be pip-installed for best clustering — the LBP-u2 fallback works without it but with reduced discrimination. (3) eps=0.38 over-splits one identity (6 clusters from 5 characters); eps=0.40 merges two distinct characters, so 6 is the better error.

## References


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